

PUBLISHED: 20-OCT-2016

2015.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - TURBOCHARGER - TDV8 4.4L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the turbocharger and operation, refer to the relevant Description and Operation section of the Workshop Manual.

INSPECTION AND VERIFICATION



WARNINGS:

- The following tests may involve working in close proximity to hot components. Make sure adequate protection is used. Failure to follow this instruction may result in personal injury.
- The turbocharger can continue to rotate after the engine has stopped. Do not attempt to check the turbocharger until one minute has elapsed since the engine was switched off. Failure to follow this instruction may result in personal injury.



CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
■ Intake air system ■ Hose(s)/hose connections ■ Turbochargers ■ General engine condition.	■ Circuit(s) ■ Electrical connections and harnesses ■ Turbocharger actuators ■ Engine Control Module (ECM)

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Poor performance (off-boost)	■ Low /Contaminated fuel ■ Restricted intake air system ■ General engine condition ■ ECM failure ■ Turbocharger damage	■ Check the fuel level and condition. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the intake air system for restriction. Check the engine condition, compressions, etc if there are indications of a mechanical fault. Check for DTCs indicating a module fault. Refer to the warranty policy and procedures manual if an ECM is suspect. ■ Using the manufacturer approved diagnostic system perform routine - Noise, vibration and harshness sensor diagnostic test - Turbocharger ■ Using the manufacturer approved diagnostic system perform routine - Inline diagnostic unit 2 diagnostic test - Turbocharger
No boost	■ Electrical connections and harnesses ■ Restricted intake air system ■ Charge air cooler restricted /leaking ■ Turbocharger actuator failure (s) ■ Turbocharger damage ■ ECM failure	■ Check the electrical connections and harnesses. Check the intake air system for restriction/leakage. Check the turbocharger actuators and circuits. Refer to the electrical guides. Check the turbochargers for wear. Disconnect the turbocharger intake and outlet pipework and turn the turbochargers by hand. Any roughness indicates a fault. Check any up and down movement in the turbocharger shafts. Excessive movement indicates a fault. If in doubt, compare the suspect unit with a new turbocharger. Check for DTCs indicating an actuator or module fault. Refer to the warranty policy and procedures manual if an ECM is suspect. ■ Using the manufacturer approved diagnostic system perform routine - Noise, vibration and harshness sensor diagnostic test - Turbocharger ■ Using the manufacturer approved diagnostic system perform routine - Inline diagnostic unit 2 diagnostic test - Turbocharger
No boost /excessive noise	■ Turbocharger damage	■ Disconnect the turbocharger intake and outlet pipework and turn the turbochargers by hand. Any roughness indicates a fault. Check any up and down movement in the turbocharger shafts. Excessive movement indicates a fault. If in doubt, compare the suspect unit with a new turbocharger.

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| | | <ul style="list-style-type: none">▪ Using the manufacturer approved diagnostic system perform routine - Noise, vibration and harshness sensor diagnostic test - Turbocharger▪ Using the manufacturer approved diagnostic system perform routine - Inline diagnostic unit 2 diagnostic test - Turbocharger |
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DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: Diagnostic Trouble Code (DTC) Index - TDV8 4.4L Diesel, DTC: Engine Control Module (ECM) (100-00 General Information, Description and Operation).